

EPFL CIVIL-127, Lab 7

March 31, 2026

7.1

Write a function `count_patterns(i: Iterable[int]) -> tuple[int, int, int]` which answers the following questions. Given an iterable (i.e. the result of calling `range()` or a list), return a tuple with 3 elements:

1. The first element of the tuple tells how many numbers have three consecutive digits. E.g.
 - 122234 should be counted. It has three consecutive digits.
 - 9322 should not be counted.
 - Note: 111222 should be counted only once.
2. The second element of the tuple tells how many numbers have all different digits. E.g.
 - 92103 should be counted, it has no repeated digit.
 - 71479 should not be counted. It has a repeated digit.
3. The third element of the tuple tells how many numbers have a pattern of alternating greater/smaller digits. E.g.
 - 285609 should be counted (because $2 < 8 > 5 < 6 > 0 < 9$)
 - 638342 should be counted (because $6 > 3 < 8 > 3 < 4 > 2$)
 - 842305 should not be counted (because $8 > 4 > 2 < 3 > 0 < 5$ doesn't alternate $<$ and $>$)
 - 285600 should not be counted (because $2 < 8 > 5 < 6 > 0 = 0$ also doesn't alternate $<$ and $>$)

You can check that your program works by checking the result for `range(1000, 10000)`. You should get (171, 4536, 3105)

7.2 (optional)

Implement a function `my_mod_exp(base: int, exp: int, mod: int) -> int` which computes $(\text{base}^{\text{exp}}) \% \text{mod}$

Use `my_mod_exp` to find the last 10 digits of:

$$9^{8765432}$$

Hint 1:



Hint 2:

